A PRESHEAF ON THE BIG H SITE IS A SHEAF IFF IT IS A ZARISKI SHEAF, SATISFIES THE EQUALIZER CONDITION FOR SURJECTIVE FLAT PROPER FINITELY PRESENTED MAPS TO AFFINE TARGETS, AND TURNS BOTH ALMOST BLOWUP SQUARES INTO CARTESIAN DIAGRAMS

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file flat.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-flat/retained-preamble.tex), a verbatim excerpt of the native Xy-pic setup and of the native theorem declarations and mathematical macros, followed by the project own definitions of the reference, slogan and history display environments (verbatim package plus three newenvironment lines) that the retained file carries at its end and that the project itself typesets as comments. No macro was invented or redefined, and the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. The counted statement and proof contain no cross reference, no citation and no included file, so the delivered body is self contained.

Editorial note. Canonical location of the counted proposition: Stacks tag 0EVI (sources/stacks-flat/pinned-original-source.tex line 11501).

Allowed standard prerequisites (editorial).

- Characterisation of h sheaves by the equalizer condition and the two almost blowup squares. Same chapter, Proposition labelled proposition-check-h, cited at proof lines 11521-11523. Reduces the sheaf condition to the cartesian property of an almost blow up square over an affine X.
- The two standard almost blowup squares and their base change behaviour. Same chapter, Examples labelled example-one-generator and example-two-generators, invoked in the statement and at proof lines 11562-11564 and 11586-11588; Lemma labelled lemma-base-change-almost-blow-up, cited at proof lines 11600-11602. Provides the explicit squares to which assumptions (3) and (4) apply, and their base changes (I) and (II).
- Uniqueness of the almost blow up square up to admissible blowups. Same chapter, Lemma labelled lemma-almost-blow-up-unique, cited at proof lines 11566-11570, and Lemma labelled lemma-almost-blow-up-square, cited at proof lines 11578-11580. Shows that whether F(X) is the fibre product does not depend on the chosen almost blow up square.

Source reference. Original definition 0EU1: sources/stacks-flat/pinned-original-source.tex, source lines 10504–10504. Same-chapter pointer into the author environment definition at source line 10504; invoked at proof line None of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

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Source reference. Original section 0EVG: sources/stacks-flat/pinned-original-source.tex, source lines 11474–11474. Same-chapter pointer into the author environment example at source line 11474; invoked at proof line None of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original section 0EVH: sources/stacks-flat/pinned-original-source.tex, source lines 11488–11488. Same-chapter pointer into the author environment example at source line 11488; invoked at proof line 11611 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original proposition 0EVF: sources/stacks-flat/pinned-original-source.tex, source lines 11292–11292. Same-chapter pointer into the author environment proposition at source line 11292; invoked at proof line 11522 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original equation 0EV6: sources/stacks-flat/pinned-original-source.tex, source lines 10957–10957. Same-chapter pointer into the author environment equation at source line 10957; invoked at proof line 11523 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0EVB: sources/stacks-flat/pinned-original-source.tex, source lines 11122–11122. Same-chapter pointer into the author environment lemma at source line 11122; invoked at proof line 11577 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0EVA: sources/stacks-flat/pinned-original-source.tex, source lines 11087–11087. Same-chapter pointer into the author environment lemma at source line 11087; invoked at proof line 11600 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0EV7: sources/stacks-flat/pinned-original-source.tex, source lines 10987–10987. Same-chapter pointer into the author environment lemma at source line 10987; invoked at proof line 11628 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Lemma 0.1. *Let $\mathcal{F}$ be a presheaf on one of the sites $(Sch/S)_h$ constructed in Definition 0EU1. Then $\mathcal{F}$ is a sheaf if and only if the following conditions are satisfied*

1. *$\mathcal{F}$ is a sheaf for the Zariski topology,*
2. *given a morphism $f : X \to Y$ of $(Sch/S)_h$ with Y affine and f surjective, flat, proper, and of finite presentation, then $\mathcal{F}(Y)$ is the equalizer of the two maps $\mathcal{F}(X) \to \mathcal{F}(X \times_Y X)$,*
3. *$\mathcal{F}$ turns an almost blow up square as in Example 0EVG in the category $(Sch/S)_h$ into a cartesian diagram of sets, and*
4. *$\mathcal{F}$ turns an almost blow up square as in Example 0EVH in the category $(Sch/S)_h$ into a cartesian diagram of sets.*

Proof. By Proposition 0EVF it suffices to show that given an almost blow up square (0EV6) with X affine in the category $(Sch/S)_h$ the diagram

$$\begin{array}{ccc} \mathcal{F}(E) & \longleftarrow & \mathcal{F}(X') \\ \uparrow & & \uparrow \\ \mathcal{F}(Z) & \longleftarrow & \mathcal{F}(X) \end{array}$$

is cartesian in the category of sets. The rough idea of the proof is to dominate the morphism by other almost blowup squares to which we can apply assumptions (3) and (4) locally.

Suppose we have an almost blow up square (0EV6) in the category $(Sch/S)_h$, an open covering $X = \bigcup U_i$, and open coverings $U_i \cap U_j = \bigcup U_{ijk}$ such that the diagrams

$$\begin{array}{ccc} \mathcal{F}(E \cap b^{-1}(U_i)) & \longleftarrow & \mathcal{F}(b^{-1}(U_i)) \\ \uparrow & & \uparrow \\ \mathcal{F}(Z \cap U_i) & \longleftarrow & \mathcal{F}(U_i) \end{array} \quad \text{and} \quad \begin{array}{ccc} \mathcal{F}(E \cap b^{-1}(U_{ijk})) & \longleftarrow & \mathcal{F}(b^{-1}(U_{ijk})) \\ \uparrow & & \uparrow \\ \mathcal{F}(Z \cap U_{ijk}) & \longleftarrow & \mathcal{F}(U_{ijk}) \end{array}$$

are cartesian, then the same is true for

$$\begin{array}{ccc} \mathcal{F}(E) & \longleftarrow & \mathcal{F}(X') \\ \uparrow & & \uparrow \\ \mathcal{F}(Z) & \longleftarrow & \mathcal{F}(X) \end{array}$$

This follows as $\mathcal{F}$ is a sheaf in the Zariski topology.

In particular, if we have a blow up square (0EV6) such that $b : X' \to X$ is a closed immersion and Z is a locally principal closed subscheme, then we see that $\mathcal{F}(X) = \mathcal{F}(X') \times_{\mathcal{F}(E)} \mathcal{F}(Z)$. Namely, affine locally on X we obtain an almost blow up square as in (3).

Let $Z \subset X$, $E_k \subset X'_k \to X$, $E \subset X' \to X$, and $i_k : X' \to X'_k$ be as in the statement of Lemma 0EVB. Then

$$\begin{array}{ccc} E & \longrightarrow & X' \\ \downarrow & & \downarrow \\ E_k & \longrightarrow & X'_k \end{array}$$

is an almost blow up square of the kind discussed in the previous paragraph. Thus

$$\mathcal{F}(X'_k) = \mathcal{F}(X') \times_{\mathcal{F}(E)} \mathcal{F}(E_k)$$

for $k = 1, 2$ by the result of the previous paragraph. It follows that

$$\mathcal{F}(X) \longrightarrow \mathcal{F}(X'_k) \times_{\mathcal{F}(E_k)} \mathcal{F}(Z)$$

is bijective for $k = 1$ if and only if it is bijective for $k = 2$. Thus given a closed immersion $Z \to X$ of finite presentation with X quasi-compact and quasi-separated, whether or not $\mathcal{F}(X) = \mathcal{F}(X') \times_{\mathcal{F}(E)} \mathcal{F}(Z)$ is independent of the choice of the almost blow up square (0EV6) one chooses. (Moreover, by Lemma 0EVA there does indeed exist an almost blow up square for $Z \subset X$.)

Finally, consider an affine object X of $(Sch/S)_h$ and a closed immersion $Z \to X$ of finite presentation. We will prove the desired property for the pair (X, Z) by induction on the number of generators r for the ideal defining Z in X. If the number

of generators is ≤ 2, then we can choose our almost blow up square as in Example 0EVH and we conclude by assumption (4).

Induction step. Suppose $X = \mathrm{Spec}(A)$ and $Z = \mathrm{Spec}(A/(f_1, \ldots, f_r))$ with $r > 2$. Choose a blow up square (0EV6) for the pair (X, Z). Set $Z_1 = \mathrm{Spec}(A/(f_1, f_2))$ and let

$$\begin{array}{ccc} E_1 & \longrightarrow & Y \\ \downarrow & & \downarrow \\ Z_1 & \longrightarrow & X \end{array}$$

be the almost blow up square constructed in Example 0EVH. By Lemma 0EV7 the base changes

$$(I) \quad \begin{array}{ccc} Y \times_X E & \longrightarrow & Y \times_X X' \\ \downarrow & & \downarrow \\ Y \times_X Z & \longrightarrow & Y \end{array} \quad \text{and} \quad (II) \quad \begin{array}{ccc} E & \longrightarrow & Z_1 \times_X X' \\ \downarrow & & \downarrow \\ Z & \longrightarrow & Z_1 \end{array}$$

are almost blow up squares. The ideal of Z in Z_1 is generated by $r - 2$ elements. The ideal of $Y \times_X Z$ is generated by the pullbacks of $f_1, \ldots, f_r$ to Y. Locally on Y the ideal generated by f_1, f_2 can be generated by one element, thus $Y \times_X Z$ is affine locally on Y cut out by at most $r - 1$ elements. By induction hypotheses and the discussion above

$$\mathcal{F}(Y) = \mathcal{F}(Y \times_X X') \times_{\mathcal{F}(Y \times_X E)} \mathcal{F}(Y \times_X Z)$$

and

$$\mathcal{F}(Z_1) = \mathcal{F}(Z_1 \times_X X') \times_{\mathcal{F}(E)} \mathcal{F}(Z)$$

By assumption (4) we have

$$\mathcal{F}(X) = \mathcal{F}(Y) \times_{\mathcal{F}(E_1)} \mathcal{F}(Z_1)$$

Now suppose we have a pair (s', t) with $s' \in \mathcal{F}(X')$ and $t \in \mathcal{F}(Z)$ with same restriction in $\mathcal{F}(E)$. Then $(s'|Z_1 \times_X X', t)$ are the image of a unique element $t_1 \in \mathcal{F}(Z_1)$. Similarly, $(s'|_{Y \times_X X'}, t|_{Y \times_X Z})$ are the image of a unique element $s_Y \in \mathcal{F}(Y)$. We claim that s_Y and t_1 restrict to the same element of $\mathcal{F}(E_1)$. This is true because the almost blow up square

$$\begin{array}{ccc} E_1 \times_X E & \longrightarrow & E_1 \times_X X' \\ \downarrow & & \downarrow \\ E_1 \times_X Z & \longrightarrow & E_1 \end{array}$$

is the base change of almost blow up square (I) via $E_1 \to Y$ and the base change of almost blow up square (II) via $E_1 \to Z_1$ and because the pairs of sections used to construct s_Y and t_1 match. Thus by the third fibre product equality we see that there is a unique $s \in \mathcal{F}(X)$ mapping to s_Y in $\mathcal{F}(Y)$ and to t_1 in $\mathcal{F}(Z)$. We omit the verification that s maps to s' in $\mathcal{F}(X')$ and to t in $\mathcal{F}(Z)$; hint: use uniqueness of s just constructed and work affine locally. $\square$